

Root Words Cheat Sheet IV — The Greenhouse Effect & Radiative Physics

This sheet leans on the Stefan–Boltzmann and Wien vocabulary from sheet one, then adds the Latin layer specific to radiative transfer: reflecting, absorbing, emitting, and the load-bearing sensible/latent heat pair, which is exactly the kind of etymology that tells you the physics rather than just labelling it.

Sensible vs Latent Heat — the load-bearing pair

sensible (Latin *sentire*, "to feel"): heat you can *sense* — it changes the temperature, so a thermometer detects it directly.

latent (Latin *latere*, "to lie hidden"): heat that is *hidden* — it goes into a phase change (melting, boiling) rather than raising the temperature, so a thermometer shows nothing while it's being absorbed or released.

The two names tell you, without any further explanation, which kind of heat you can detect and which kind is doing invisible work.

RADI- *ray, spoke* (Latin *radius*)

radiation, radiance, irradiance, radiant, radiator, radial

ALB- *white* (Latin *albus*)

albedo — literally "whiteness," the fraction of incoming radiation reflected back

EMIT-/EMISSION- *send out* (Latin *emit-tere*)

emissivity, emit, emission, emitter

ABSORB-/SORPT- *swallow, suck in* (Latin *absorbere*)

absorb, absorption, absorptivity

RE- *back, again* (Latin — callback to sheet one)

reflect, reflectivity, re-emit, re-radiate
Same root as reflection/refraction on sheet one, doing thermal rather than optical work.

RESON- *sound again* (Latin *re- + sonare*, to sound)

resonance, resonant absorption — the vibrational-mode explanation for CO₂ absorbing infrared is a literal echo of the acoustic meaning

-SPHERE *ball* (Greek *sphaira*)

atmosphere, troposphere, stratosphere, biosphere, cryosphere
tropo- = turning (the mixing layer); *strato-* = layered (the stratified layer above it)

OPAC-/TRANSP- *shady, dark / across* (Latin)

opacity, opaque, transparent, transmit, transmittance

FLUX-/FLU- *flow* (Latin *fluere*)

flux, influx, fluid — same root family as "fluent"

INSOL- *sun* (Latin *in-* + *solare*)

insolation

Trap: easily confused with "insulation," a different root entirely (insula, "island" — to insulate is to make into an island, cut off).

ANTHROPO- *human* (Greek *anthropos*)

anthropogenic

CONVECT-/CONDUCT- *carry together / lead together* (Latin *vehere, ducere*)

convection, conduction — the two non-radiative transfer mechanisms, worth pairing directly

FORC-/FEED- plain English, not a Latin/Greek fossil

radiative forcing, feedback — a reminder that not every technical term is inherited; some are just precise, repurposed English

Test Yourself

1. Explain, using the roots, why "sensible heat" and "latent heat" are named the way they are, and state which one a thermometer can detect.
2. *Albedo* comes from the Latin for "white." Explain why this is an apt name for the quantity it describes.
3. State the roots of *emissivity* and *absorptivity* and explain, in one sentence each, what a surface with emissivity/absorptivity close to 1 does.
4. *Reflect* appears on both sheet one (optics) and this sheet (radiative heat transfer). State the shared root and explain what stays the same about its meaning in both contexts.
5. *Resonance* comes from a root meaning "to sound again." Explain how this acoustic origin actually describes what happens when a CO₂ molecule absorbs infrared radiation.

6. Explain the difference between *tropo-* ("turning") and *strato-* ("layered") and why each is an accurate name for its atmospheric layer.
7. *Insolation* and *insulation* look almost identical but come from entirely different roots. State both roots and their meanings, and explain the trap.
8. *Convection* and *conduction* share the sense of "carrying" or "leading," but describe different physical processes. Explain the distinction.
9. *Radiative forcing* and *feedback* are not inherited from Latin or Greek roots the way most words on this sheet are. Explain why this matters when you're trying to guess an unfamiliar word's meaning from its roots.
10. *Anthropogenic* combines a Greek root for "human" with the *-gen* root from sheet three (producing, born of). Use both roots to explain what the word means in the context of climate change.

Answer Bank

1. *Sensible* (Latin *sentire*, "to feel/sense") describes heat that changes temperature, which a thermometer can sense directly. *Latent* (Latin *latere*, "to lie hidden") describes heat absorbed or released during a phase change, which does not change temperature and so stays hidden from a thermometer.
2. *Albedo* measures the fraction of light reflected rather than absorbed; a perfectly white surface reflects all incoming light, so "whiteness" is a natural name for a quantity that ranges from 0 (absorbs everything, black) to 1 (reflects everything, white).
3. *Emissivity* (*emittere*, "send out") describes how effectively a surface sends out radiation; a surface with emissivity near 1 emits almost as effectively as a perfect black body. *Absorptivity* (*absorbere*, "swallow, suck in") describes how effectively a surface takes in incoming radiation; a surface with absorptivity near 1 absorbs almost all incoming radiation rather than reflecting or transmitting it.
4. The shared root is *re-*, "back." In optics, light reflects back off a surface at an angle; in radiative heat transfer, absorbed or incoming radiation reflects back rather than being absorbed. In both cases the core meaning, "sent back," stays exactly the same.
5. "To sound again" describes a system re-amplifying a specific frequency it is tuned to. A CO₂ molecule has vibrational modes tuned to particular infrared frequencies; when infrared radiation at a matching frequency arrives, the molecule absorbs it strongly because the incoming radiation "sounds again" with the molecule's own natural vibration, exactly as an acoustic resonance amplifies a matching sound frequency.
6. *Tropo-* ("turning") names the troposphere because it is the layer where convection actively turns and mixes air vertically. *Strato-* ("layered") names the stratosphere because temperature increases with altitude there, which suppresses vertical mixing and keeps the air in stable, distinct layers instead.
7. *Insolation* comes from *in-* + *solare* ("sun"), meaning incoming solar radiation. *Insulation* comes from *insula* ("island"), meaning material that cuts a system off, like an island, from heat exchange with its surroundings. The trap is that the words look and sound similar but describe opposite-feeling concepts: one is about receiving solar energy, the other about blocking heat flow entirely.
8. Convection (*vehere*, "to carry") describes heat carried by the bulk movement of a fluid (rising warm air, sinking cool air). Conduction (*ducere*, "to lead") describes heat led directly through a material via particle-to-particle vibration, with no bulk movement of the material itself.
9. Because these are ordinary English words given precise technical meanings, their etymology gives no extra clue about the science; root-based reasoning stops working for genuinely modern coinages, and you have to learn the definition directly rather than decode it.
10. *Anthropo-* ("human") + *-genic* ("producing, born of") together mean "produced by humans." Anthropogenic climate change is change to the climate system caused by human activity, as distinct from natural forcing mechanisms like solar variation or volcanic activity.

Matching Quiz: Prefix to Meaning

Draw a line (or write the matching letter) connecting each prefix/root on the left to its correct meaning on the right. Meanings are deliberately shuffled and there are more meanings than prefixes, so some will be left over.

Prefix/Root		Meaning
1. RADI-	_____	A. flow
2. ALB-	_____	B. ray, spoke
3. EMIT-/EMISS-	_____	C. white
4. ABSORB-	_____	D. send out
5. RESON-	_____	E. swallow, suck in
6. FLUX-/FLU-	_____	F. sound again
7. SENS-	_____	G. lie hidden
8. LATE-	_____	H. human
9. ANTHROPO-	_____	I. to feel
10. OPAC-	_____	J. shady, dark
		K. across
		L. ball

Bonus: pick one leftover meaning from the right-hand column and name a word from any of the four sheets that uses it.

Tricky Words: Construct Your Own Definition

1. **Pyranometer** (*pyr-* = fire, *ano-* = up, *-meter* = measure)

Your definition:

2. **Diurnal** (Latin *dies* = day)

Your definition:

3. **Hygroscopic** (*hygro-* = moist, *-scopic* = observing)

Your definition:

4. **Solstice** (*sol* = sun, *sistere* = to stand still)

Your definition:

5. **Perihelion** (*peri-* = near, around, *helios* = sun)

Your definition:

6. **Aphelion** (*apo-* = away from, *helios* = sun)

Your definition: